

———1. Which of the skin's many functions are partly or completely provided by components of the **dermis**?

- a) protection from dehydration b) regulation of blood pressure c)
synthesis of melanin
d) protection from ultraviolet radiation e) synthesis of keratin

———2. Which connective tissue layer lies **closest to the epidermis**:

- a. reticular layer of dermis. b. papillary layer of dermis. c. hypodermis. d. lamina propria.
e.
mucosa.

———3. The majority of cells comprising the **epidermis** are called:

- a. keratinocytes b. melanocytes. c. Langerhans cells. d. Merkel cells.
e. fibroblasts.

———4. Which of the following is **NOT** an "**epidermal appendage**" (i.e., NOT comprised of epithelial tissue)?

- a.sebaceous gland. b. hair follicle. c. duct
of sweat gland. d. secretory portion of sweat gland. e. arrector pili.

———6. Which of the following **epidermal layers** is most likely to contain both **desmosomes** &

hemidesmosomes?

- a) stratum basale b) stratum corneum c)
stratum granulosum
d) stratum lucidum e) stratum spinosum

———7. **Arrector pili muscles.**

- a) are composed of bundles of myoepithelial cells. b) help push sebum onto
the skin surface.
c) have no effect on hair position in humans. d) contain capillary networks that nourish
growing hairs.
e) attach directly to the external root sheath by desmosomes.

xx. The most common cells in dermis are:

- a. Keratinocytes. b. Melanocytes. c. Langerhans
cells.
d. Fibroblasts. e. Merkel cells.

6. In the **connective tissue**, which statement is not correct?

- a. Costal cartilage are made up of hyaline cartilage. b. Osteon is unmineralized bone
matrix.

c. The collagen in elastic cartilage matrix is mainly of type II. d. Osteoclast is involved in bone resorption.

e. Sharpey's fibers are collagen fibers that extend into a bone at an angle.

7. Which type of cartilage forms the **malleus, incus** and **stapes** of the **middle ear**?

a. Hyaline cartilage b. Elastic cartilage c. Fibrocartilage d. All of them e. None of them

18. Concerning the development of the **pharyngeal apparatus**, which statement is **false**?

a) Pharyngeal groove 1 gives rise to the external acoustic meatus.

b) Glossopharyngeal nerve is the 3rd pharyngeal arch cranial nerve.

c) The inferior parathyroid gland is derived from the 3rd pharyngeal pouch.

d) Musculature of the 1st pharyngeal arch forms the muscles of mastication.

e) Ventral end of the 2nd arch cartilage ossifies to form the stapes of the middle ear and the styloid process of the temporal bone.

——— **16.** Which **connective tissue component** is characterized by high tensile strength and flexibility?

a. collagen b. elastin c. glycosaminoglycan d. fibroblasts
e. lipid

——— 17. Cells which secrete the components of **bone matrix** are called:

a. fibroblasts. b. chondroblasts. c. osteoblasts. d. osteoclasts.
e. osteocytes.

——— **18.** With regard to **cartilage**, which statement is **false**?

a) Hyaline cartilage is characterized by an amorphous glassy matrix.

b) Fibrocartilage has thick bundles of collagen fibres in the matrix.

c) Fibrocartilage is present in regions subjected to high tension.

d) Elastic cartilage is similar to hyaline cartilage except the abundant elastic fibres it contains.

e) Fibrocartilage is used for endochondral bone ossification.

——— 19. **Bone formation** without a pre-existing cartilage model is called:

a. intramembranous bone formation. b. extramembranous bone formation.

c. endochondral bone formation. d. trabecular bone formation. e. cancellous bone formation.

——— **20.** In the **restructuring of the skeleton, bone remodeling**:

a. follows a cycle in which bone formation alternates with resorption. b. occurs exclusively in adult bone. c. occurs independently of soft tissue growth and external forces.

d. only occurs at the epiphyses of long bones. e. results in the formation of cement lines.

—————21. **Elastic cartilage**,

- a) is found in the annulus fibrosus and intervertebral disks.
- b) is the primary skeletal tissue in the fetus.
- c) has no identifiable perichondrium.
- d) the collagen in its matrix is mainly type II.
- e) is the most widely distributed cartilage type in the body.

—————22. Which of the following cells **does not** belong to the **mononucleated phagocytic system**?

- a) Merckel's cells
- b) Dust cells
- c) Langerhan's cells
- d) Kupffer cells
- e) Cells of Ito

—————23. Which of the following statements is **correct** regarding the **bone**?

- a) Primary bone has a higher mineral content than the secondary bone.
- b) Sharpey's fibers are calcified collagen bundles that run in concentric manner.

—————71. The **glands most closely associated with hair follicles** are:

- a. eccrine sweat glands.
- b. apocrine sweat glands.
- c. sebaceous glands.
- d. lymphatic glands.
- e. submucosal glands.

—————72. Which of the **skin's many functions** are partly or completely provided by components of the **dermis**?

- a) protection from dehydration
- b) regulation of blood pressure
- c) protection from ultraviolet radiation
- d) synthesis of keratine
- e) synthesis of melanin

—————73. Which of the following **epidermal layers** is most likely to contain both **desmosomes&hemidesmosomes**?

- a) stratum basale
- b) stratum corneum
- c) stratum granulosum
- d) stratum lucidum
- e) stratum spinosum

—————74. Which of the following is (are) often **absent in thick skin** but typically present in **thin skin**?

- a) arrector pili muscles
- b) Meissner's corpuscles
- c) sweat glands
- d) stratum basale
- e) stratum spinosum

—————75. **Cells of epidermis** that form the stratified squamous keratinized epithelium are:

- a) Keratinocytes
- b) Melanocytes
- c) Langerhan's cells
- d) Merckel's cells
- e) Chondrocyte

—————76. **Sebaceous gland** is a type of:

- a) Compound tubular gland.
- b) Compound alveolar gland.
- c) Simple straight tubulargland.
- d) Simple branched alveolar gland.
- e) Simple coiled tubulargland

16. The **cranial half of the 1st cervical sclerotome** gives rise to:

- a) Upper part of body of the 1st cervical vertebra.
- b) Lower part of body of the 1st cervical vertebra.
- c) Whole body of the 1st cervical vertebra.
- d) Part of base of occipital bone.
- e) Part of roof of the cranial cavity.

17. Which of the following **muscles** develop from the **epimeres**?

- a) Intercostal muscles.
- b) The strap muscles of the neck.
- c) Rectus abdominis.
- d) Quadratus lumborum.
- e) Extensor muscles of the back.

18. Failure of **fusion of the neural arches of the developing vertebrae** could lead to:

- a) Achondroplasia
- b) Acromegally
- c) Spina bifida
- d) Brachycephaly
- e) Amelia

9. Structures derived from endoderm of the **second pharyngeal pouch** include:

- a) Thymus gland
- b) Palatine tonsil
- c) Thyroid gland
- d) Parafoollicular cells of thyroid gland
- e) All of the above

11. With regard to **cartilage**, which statement is **false**?

- a) Elastic cartilage is similar to hyaline cartilage except the abundant elastic fibres it contains.
- b) Hyaline cartilage is characterized by an amorphous glassy matrix.
- c) Fibrocartilage has thick bundles of collagen fibres in the matrix.
- d) Fibrocartilage is present in regions subjected to high tension.
- e) Fibrocartilage is used for endochondral bone ossification.

12. Elastic cartilage,

- a) is found in the annulus fibrosus and intervertebral disks.
- b) the collagen in its matrix is mainly type II.
- c) is the primary skeletal tissue in the fetus.
- d) has no identifiable perichondrium.
- e) is the most widely distributed cartilage type in the body.

13. With regard to **bone**, which statement is **false**?

- a) Compact (dense) bone is characterized by the presence of osteons.
- b) Bone formation without a pre-existing cartilage model is called intramembranous bone formation.
- c) In the restructuring of the skeleton, bone remodeling follows a cycle in which bone formation alternates with resorption.

d) Osteoblasts secrete the components of bone matrix. e) Osteocytes are involved in bone resorption.

_____14. Which of the following statements is **correct** regarding the **bone**?

- a) Primary bone has a higher mineral content than the secondary bone.
- b) Sharpey's fibers are calcified collagen bundles that run in concentric manner.
- c) Bone contains matrix with more inorganic components than organic component in its dry weight.
- d) Osteoclasts have processes that run in their canaliculi.
- e) Osteoblast is derived from blood monocytes

_____21. With regard to the **development of the musculoskeletal system**,

- a) The muscular system develops from neuroectoderm.
- b) The upper limbs rotate laterally through 90° on their longitudinal axes.
- c) Premature closure of the frontal (metopic) suture results in scaphocephaly.
- d) Sclerotomes of the somites form the viscerocranium.
- e) Xiphoid process develops during the 6th week.

_____22. With regard to the **pharyngeal apparatus**, which statement is **false**?

- a) The fourth inferior pouch forms the ultimopharyngeal body.
- b) The second aortic arch forms the hyoid and stapedial arteries.
- c) The second pharyngeal/branchial arch is associated with muscles of facial expression.
- d) The third superior pouch forms the inferior parathyroid_____e)
- None of the above.

_____23. Structure that is **not** developed in the **third branchial arch** include:

- a) Lesser cornu of hyoid bone
- b) Body of the hyoid bone
- c) Glossopharyngeal nerve
- d) Common carotid artery
- e) Stylopharyngeus muscle

_____24. Which of the following structures develop from **endoderm of the second pharyngeal pouch**?

- a) Thymus gland
- b) Palatine tonsil
- c) Thyroid gland
- d) Superior parathyroid gland
- e) Parafoallicular cells of thyroid gland

_____31. Concerning the development of the **Integumentary system**, which statement is **false**?

- a) Up to the 5th week of development, the embryonic skin is only a single layer of ectoderm.
- b) Leprosy results from excessive keratinization of the skin
- c) Melanocytes are derived from neural crest.
- d) Dermis is derived from lateral plate mesoderm.
- e) None of the above.

_____ 27. With regard to the **development of the musculoskeletal system**, which statement is **true**?

- a) The muscular system develops from neuroectoderm.
- b) The upper limbs rotate laterally through 90° on their longitudinal axes.
- c) Premature closure of the frontal (metopic) suture results in scaphocephaly.
- d) Sclerotomes of the somites form the viscerocranium.
- e) Xiphoid process develops during the 6th week.

2. Which connective tissue cell type is functionally related to the basophiles?

- a) Fibroblast b) Myofibroblast c) Histiocyte d) Plasma cell e) Mast cell

3. With regard to cartilage, which statement is **false**?

- a) Hyaline cartilage is characterized by an amorphous glassy matrix.
- b) Fibrocartilage has thick bundles of collagen fibres in the matrix.
- c) Fibrocartilage is present in regions subjected to high tension.
- d) Elastic cartilage is similar to hyaline cartilage except the abundant elastic fibres it contains.
- e) Fibrocartilage is used for endochondral bone ossification.

4. Which type of bone cells are involved in bone resorption?

- a) Osteoclast b) Osteocyte c) Osteoblast d) All of the above e) None of the above

8. Cells of epidermis that form the stratified squamous keratinized epithelium are:

- a) Keratinocytes b) Melanocytes c) Langerhan's cells d) Merckel's cells e) Chondrocyte

9. Sebaceous gland is a type of:

- a) Compound tubular gland.
- b) Compound alveolar gland.
- c) Simple straight tubular gland.
- d) Simple branched alveolar gland.
- e) Simple coiled tubular gland.

Bone formation without a pre-existing cartilage model is called:

- a. **intramembranous bone formation.**
- b. **extramembranous bone formation.**
- c. **endochondral bone formation.**
- d. **trabecular bone formation.**
- e. **cancellous bone formation.**

The activity of osteoclasts is inhibited, decreasing blood calcium levels, by secretion from the:

- a. **thyroid follicles.**
- b. **adrenal cortex.**
- c. **parathyroid.**
- d. **thyroid parafollicular cells (C cells).**
- e. **neurohypophysis.**

In the restructuring of the skeleton, bone remodeling:

- a. follows a cycle in which bone formation alternates with resorption.
- b. occurs exclusively in adult bone.
- c. occurs independently of soft tissue growth and external forces.
- d. only occurs at the epiphyses of long bones.
- e. results in the formation of cement lines.

In compact bone, osteocytes which are not adjacent to capillaries receive or exchange nutrients and metabolites by means of:

- a. processes contacting other osteocytes along canaliculi.
 - b. lymph drainage through central (Haversian) canals.
 - c. diffusion through the bony matrix.
 - d. fenestrations in medullary vessels.
 - e. active transport across the endosteum.
- f. ———7. The distance between two **Z-lines in skeletal muscle** is equivalent to the length of the:
- g. a. A band b. H band c. I band d. M line e. sarcomere

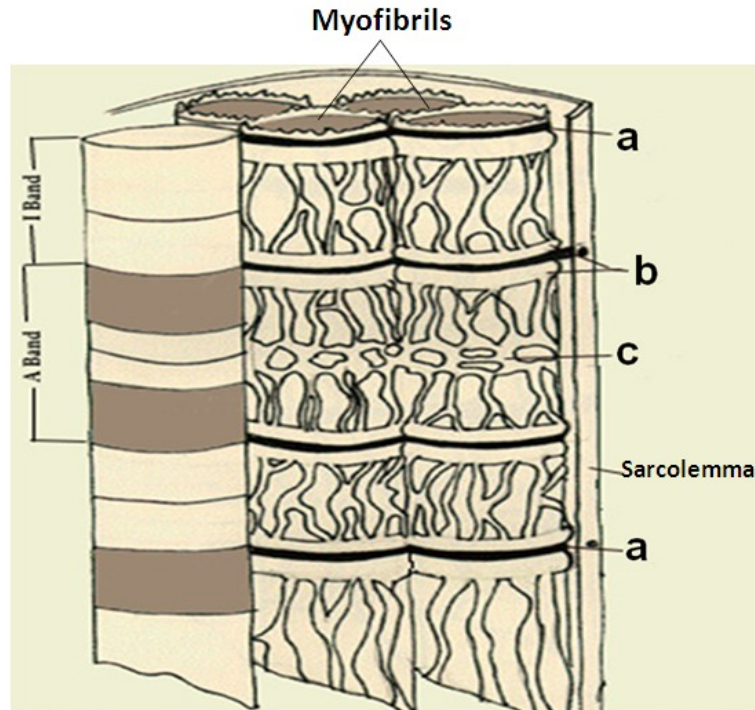
1. The sarcoplasmic reticulum of skeletal muscle cells are made by:

- a) Golgi apparatus b) Polyribosomes c) Rough endoplasmic reticulum.
- d) Smooth endoplasmic reticulum. e) Mitochondria.

2. Regarding the striated muscle cells,

- a) The striations of the light microscopy are manifested because of extensive plasma membrane foldings
- b) The location of triads in the myofibrils of skeletal muscle cells is identical in all animals.
- c) Both skeletal and cardiac muscle cells show regenerative capacity from the surrounding pericytes if damaged.
- d) Intercalated discs of cardiac muscle are made by cell junctions.
- e) All skeletal muscles are found attached to bones.

3-6. Answer questions 8-11 from Figure 1 below representing a muscle cell.



3. Which type of muscle cell does the figure represent?

- a) Skeletal muscle b) Cardiac muscle c) Smooth muscle d) a & b e) All

4. The regions indicated by letter "a" are:

- a) Terminal cisternae b) Transverse tubule c) Sarcotubules
d) Intercalated duct e) Diad

5. The regions indicated by letter "b" are:

- a) Terminal cisternae b) Transverse tubule c) Sarcotubules
d) Endomysium e) Diad

6. The region indicated by letter "c" is:

- a) Terminal cisternae b) Transverse tubule c) Sarcotubules
d) Mitochondria e) Diad

10. Which of the following structures develop from the costal processes of the sclerotomes?

- a) The transverse processes b) Foramina transversaria
c) Lateral mass or alae of the sacrum d) All of the above e) None of the above

11. The cranial half of the 1st cervical sclerotome:

- a) Gives rise to the upper part of body of the 1st cervical vertebra.
b) Gives rise to the lower part of body of the 1st cervical vertebra.
c) Gives rise to the whole body of the 1st cervical vertebra.
d) Participates in the formation of base of occipital bone. e) Disappears

12. The body of 1st thoracic vertebra is formed by:

- a) The caudal half of the 8th cervical sclerotome alone.
b) The cranial half of the 1st thoracic sclerotome alone.

- c) The caudal half of the 1st thoracic sclerotome alone.
- d) Fusion of caudal half of the 1st thoracic with cranial half of the 2nd thoracic sclerotomes.
- e) Fusion of caudal half of the 8th cervical with cranial half of the 1st thoracic sclerotomes.

13. The spinal nerves, in relation to their corresponding sclerotomes, were initially located:
- a) Above their corresponding sclerotomes.
 - b) Below their corresponding sclerotomes.
 - c) At the level of succeeding sclerotome.
 - d) At the level of preceding sclerotome.
 - e) At the level of middle of corresponding sclerotomes.

14. Bones of the limbs develop from:
- a) Somatic layer of lateral mesoderm.
 - b) Visceral layer of lateral mesoderm.
 - c) Sclerotomes.
 - d) Intermediate mesoderm.
 - e) Neural crest.

15. Which of the following statements is correct with regard to the rotation of the limbs during the 7th week?

- a) Lower limb rotates 180° medially, while upper limb rotates 180° laterally.
- b) Lower limb rotates 180° laterally, while upper limb rotates 180° medially.
- c) Lower limb rotates 90° laterally, while upper limb rotates 90° medially.
- d) Lower limb rotates 90° medially, while upper limb rotates 90° laterally.
- e) Lower limb rotates 90° laterally, while upper limb rotates 180° medially.

16. The dorsal somatic mesodermal condensations in the limb bud give rise to which types of muscles?

- a) Extensors and pronators.
- b) Extensors and abductors.
- c) Extensors and flexors.
- d) Flexors and pronators.
- e) Flexors and abductors.

17. Which of the following statements is true regarding the hypomeric and epimeric divisions of the myotomes?

- a) Epimers are larger and located dorsally as compared to the hypomers.
- b) Epimers are smaller and located dorsally as compared to the hypomers.
- c) Epimers are innervated by the dorsal branches of the ventral primary rami of the spinal nerves.
- d) Hypomers give rise to the extensors of the back.
- e) None of the above.

18. Cardiac muscles develop from:

- a) Somatic layer of lateral mesoderm.
- b) Splanchnic layer of lateral mesoderm.
- c) Intermediate mesoderm.
- d) Both splanchnic and somatic mesoderm.
- e) Myotomes.

19. Which of the following is a condition caused by early closure of epiphyseal plates of long bones leading to dwarfism?

- a) Acrania
- b) Acromegally
- c) Achondroplasia
- d) Klippel-Feil syndrome
- e) None of the above

20. Which of the following bones **does not** developed in the 1st arch?

- a) Incus
- b) Malleus
- c) Maxilla
- d) Mandible
- e) Styloid process of temporal bone

21. Structure that is not developed in the third branchial arch includes:

- a) Lesser cornu of hyoid bone
- b) Glossopharyngeal nerve
- c) Common carotid artery
- d) Stylopharyngeus muscle
- e) Lower part of body of hyoid bone

3. Which of the following structures in the skin develops from a different germ layer than the others?

- a) Hairs
- b) Sweat glands
- c) Arrector pili muscle
- d) Nails
- e) Apocrine sweat glands

45. What type of tissue makes up the **dermis of the skin**?

- a. Mucous connective tissue
- b. Mesenchyme
- c. Loose irregular connective tissue
- d. Dense irregular connective tissue
- e. Dense regular connective tissue

____ 51. What is the **mature cell in cartilage** called?

- a. Chondrocyte
- b. Chondroblast
- c. Osteocyte
- d. Osteoclast
- e. Bone lining cell

____ 52. Which type of **cartilage** is characterized by the presence of **chondrocytes sitting in lacunae**?

- a. Hyaline cartilage
- b. Elastic cartilage
- c. Fibrocartilage
- d. All of them
- e. None of them

____ 53. Which **type of cartilage** forms the hammer, anvil and stirrup?

- a. Hyaline cartilage
- b. Elastic cartilage
- c. Fibrocartilage
- d. All of the above
- e. None of the above

____ 54. Which type of **cartilage** is highly vascular?

- a. Hyaline cartilage
- b. Elastic cartilage
- c. Fibrocartilage
- d. All of the above
- e. None of the above

____ 55. **Hyaline cartilage** is not found in the

- a) skeleton of the fetus.
- b) epiphyseal growth plate.
- c) articular surface on bones.
- d) rings of the trachea
- e) epiglottis

____ 56. There is **no fibrocartilage** in the

- a) larynx
- b) symphysis pubis.
- c) intervertebral disc.

d) temporomandibular joint
None of them

e) All of them

a)

_____ 57. Concerning the **cartilage**, which statement is **true**?

a) Fibrocartilage is found in the epiphyses of the long bones in young children.

b) Cartilage harbors hematopoietic organs. c) The collagen in elastic cartilage matrix is mainly type IV.

d) Elastic cartilage is the primary skeletal tissue in the fetus. e) articular cartilage lacks a perichondrium.

_____ 58. Which of the following **cell types in bone** is derived from **blood monocytes**?

a) endothelial cell b) fibroblast c) osteoblast d) osteoclast
e) osteocyte.

_____ 59. **Primary bone**,

a) is also termed lamellar bone. b) may appear as either spongy or compact bone.

c) is characterized by the presence of Volkmann's canal. d) has a higher mineral content than the secondary bone.

e) is the first bone tissue to appear during both intramembranous and enchondral bone formation.

_____ 60. Which of the following **impedes the distribution of nutrients and oxygen to osteocytes**?

a) bone matrix b) canaliculi c) filopodia d) Haversian canals
e) periosteum.

_____ 61. Which of the following hormones is most responsible for increasing **osteoclast** number and activity?

a) ADH b) calcitonin c) estrogen d) parathyroid hormone
e) testosterone.

_____ 62. Select the **correct statement**.

- a) Osteocyte is involved in laying down new bone.
- b) Howship's lacuna is unmineralized bone matrix.
- c) Endosteum is attached to the surface of bone by Sharpey's fibers.
- d) Compact (dense) bone is characterized by the presence of osteons.
- e) Dense regular connective tissue is abundantly found in the bone matrix.

_____ 63. With regard to **bone morphology**, which statement is **false**?

- a. Spicules are the unmineralized tissue in spongy bone.
- b. An osteoclast is a bone resorbing cell.
- c. The osteoblast is the bone forming cell.
- d. Osteoid is unmineralized bone matrix.
- e. Trabeculae are the spicules seen with spongy bone.

_____ 64. Which type of bone has **spicules**?

- a. Immature bone
- b. Dense bone
- c. Compact bone
- d. Cancellous bone
- e. Woven bone

_____ 65. What is the primary component of **red marrow**?

- a. Hematopoietic tissue
- b. Fat
- c. Cartilage
- d. Fibrous tissue
- e. Bone

9. Which cell is an **immature bone cell**?

- a. Osteoclast
- b. Osteon
- c. Osteocyte
- d. Osteoblast
- e. Osteoid

_____ 66. What is in the **bone matrix**?

- a. Elastic fibers
- b. Collagen fibers
- c. Reticular fibers
- d. Dense irregular connective tissue
- e. Dense regular connective tissue

_____ 67. What is **immature bone**?

- a. Dense bone
- b. Woven bone
- c. Cancellous bone
- d. Compact bone
- e. Spongy bone

_____ 68. Concerning the **bone**, which statement is **false**?

- a) Osteoblasts synthesize and secrete osteon.
- b) Osteoprogenitor cells are capable of differentiating into osteoblasts.
- c) Osteoprogenitor cells are located in the periosteum and endosteum.
- d) Osteoprogenitor cells may change into chondrogenic cells at low oxygen tensions.
- e) Chondroclasts found in eroded regions of epiphyseal calcified cartilage are similar in shape to osteoclasts.

_____ 69. With regard to **bone**, which statement is **true**?

- a) Periosteum is a layer of dense connective tissue surrounding the bone.
- b) As the osteoclasts become trapped in their own calcified deposits, they are known as osteoblasts.
- c) Primary bone is remodelled and replaced by secondary bone in tooth sockets and near suture lines in skull bones.
- d) The ruffled borders of osteocytes are irregular finger-like cytoplasmic projections extending into Howship's lacunae.
- e) osteoclast activity is hormonally regulated, stimulated by calcitonin from C-cells of the thyroid and inhibited by
parathyroid hormone.